
Lab Report: Autonomous Navigation with Nav2

Course: MCT-454L Mobile Robotics

Student: Sheraz

1. Objective

The objective of this lab was to enable autonomous navigation for a TurtleBot3 in a Gazebo simulation using the Nav2 stack. This included loading a pre-built map, localizing the robot with AMCL, and executing multi-waypoint missions via a custom Python node.

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2. System Architecture & Components

The Nav2 framework used during this lab consists of several key servers:

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- **Map Server:** Loads the static occupancy grid (.pgm and .yaml).

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- **AMCL (Adaptive Monte Carlo Localization):** Estimates the robot's pose on the map using a particle filter and LiDAR scans.

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- **Planner Server:** Computes the global path from the current position to the target.

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- **Controller Server:** Follows the global path by generating local velocity commands (DWB controller).

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- **Waypoint Follower:** Manages the sequence of goals for the mission.

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3. Experimental Procedure

1. **Simulation Launch:** Launched the TurtleBot3 world in Gazebo.

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2. **Navigation Setup:** Initialized the Nav2 bringup with the map saved from Lab 5.

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3. **Localization:** Set the initial pose in RViz using the **2D Pose Estimate** tool and verified the AMCL particle cloud convergence.

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4. **Single Goal Testing:** Sent manual goals via RViz to verify path planning and obstacle avoidance.

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5. **Automated Navigation:** Developed and executed waypoint_navigator.py to automate a 5-waypoint mission.

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4. Task 2: Multi-Waypoint Mission Data

Waypoint	X (m)	Y (m)	Z (m)	Yaw (rad)	Description
1	0.5	0.0	0.0	1.0	Initial move from origin
2	1.0	0.5	0.0	1.0	Corner navigation
3	1.0	-0.5	0.0	1.0	Straight line traverse
4	0.0	0.5	0.0	0.0	Mid-map checkpoint
5	0.0	0.0	0.0	1.0	Return to origin

5. Observations & Results

- **Costmap Role:** The **Global Costmap** was used for long-distance path planning, while the **Local Costmap** allowed the robot to sense and avoid obstacles in real-time.

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- **Recovery Behaviors:** When a dynamic obstacle was placed in Gazebo, the robot attempted to:

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- *Observation (e.g., Replan a new path or rotate to find clear space).*

- **Localization Accuracy:** AMCL localization became more accurate as the robot moved, evidenced by the particle cloud shrinking around the robot model.

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6. Conclusion

This lab successfully demonstrated the integration of mapping and autonomous navigation. While SLAM (Lab 5) focuses on building a representation of an unknown environment, Navigation (Lab 7) focuses on utilizing that representation to achieve complex, multi-stage goals autonomously.
